

Mikael Elgo
Geophysicist

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Geophysical Engineering graduate from the Bandung Institute of Technology (ITB) with training in seismic data processing, seismic interpretation, and subsurface analysis. Experience with 2D and 3D seismic datasets, marine and land acquisition data, and reservoir characterization workflows. Familiar with integrating seismic results with geological and petrophysical information for subsurface studies. Background includes field operations, data quality control, and teamwork in academic and industry-related environments.

Professional Experience

Mahakarya Geo Survey
Geophysicist

Jakarta, Indonesia
October 2025 – Present

- Trained to process and quality-check marine geophysical datasets from individual offshore survey areas, including side-scan sonar, sub-bottom profiler, and marine magnetometer data, as well as 2D and 3D marine seismic datasets from two separate fields.
- Performed marine seismic data processing and quality control using ProMAX SeisSpace, developing and applying a custom processing workflow up to Prestack Depth Migration (PSDM).
- Assisted in the integration of geophysical results with bathymetric, geotechnical, and metocean datasets to support offshore site investigation studies.

Pertamina Hulu Rokan
Geophysicist Intern

Prabumulih, Indonesia
April 2025 – August 2025

- Interpreted 3D seismic data and integrated well log data (sonic, density, and petrophysical logs) using Petrel and Interactive Petrophysics (IP) to delineate thin, interbedded sandstone–shale reservoirs within the Talang Akar Formation, South Sumatra Basin.
- Performed seismic inversion methods to analyze lateral reservoir heterogeneity and depositional variations, supporting improved understanding of reservoir distribution in a mature oil field.
- Conducted detailed seismic interpretation to identify potential bypassed hydrocarbon zones, providing subsurface insights that supported field development evaluation and enhanced recovery planning.

Seismic Refraction Practicum ITB
Practicum Assistant Coordinator

Bandung, Indonesia
September 2024 – January 2025

- Led field operations and supervised 60+ students in seismic refraction survey design, data acquisition, and processing, ensuring proper data collection and strict adherence to HSE standards.
- Trained students in Refrappy software for tomography and Hagiwara processing methods, resulting in an average practicum score of 81.2 (16% above the passing grade of 70).
- Managed equipment logistics and on-site troubleshooting during field practicums, minimizing survey downtime and ensuring high-quality data output.
- Acted as a liaison between students and lecturers, providing timely updates on session progress and addressing technical challenges.

Lecturer-Led Project (LAPI ITB)
Project Assistant

Bandung, Indonesia
August 2024 – October 2024

- Prepared and validated Seismic Processing Support (SPS) files (shot, receiver, and cross-reference data) for large-scale 3D land seismic projects, supporting acquisition activities for SWATH 38–99 (Antara) and SWATH 1–38 (Balam South).
- Performed geometry quality control and initial seismic processing on 3D seismic datasets using VISTA, covering survey areas of over 4,000 km², to ensure data integrity and accurate subsurface imaging geometry.

Organizational Activities

HIMA TG "TERRA" ITB

President

Bandung, Indonesia

April 2024 – March 2025

- Directed organizational strategy and operations for a 200+ member geophysical engineering student association, doubling the number of programs (20+ initiatives) compared to the previous term.
- Coordinated 20+ academic, professional, and social events in collaboration with faculty and industry partners, boosting member engagement and skill development.
- Represented the organization in faculty and inter-university forums, forging partnerships and expanding networks for research collaborations and career opportunities.

KAT ITB 2023

Field Commander

Bandung, Indonesia

July 2023 – August 2023

- Led field operations and safety protocols for university-wide student orientation programs (OSKM and Diklat Terpusat) with over 10,000 and 2,000 participants respectively.
- Coordinated 300+ organizing committee members across security and medical divisions to ensure crowd safety, schedule adherence, and zero major incidents during high-density events.
- Ensured strict compliance with Health, Safety, and Environment (HSE) protocols during large-scale field activities, resulting in zero major incidents throughout the event.

Kaderisasi Wilayah FTTM ITB 2023

Field Commander

Bandung, Indonesia

May 2023 – Juni 2023

- Directed field operations and ensured the smooth execution of Kaderisasi Wilayah FTTM ITB activities attended by approximately 1,000 participants, managing on-site coordination and operational readiness.
- Coordinated organizing teams and logistics, including scheduling, area management, and participant flow, to maintain order and program continuity throughout the event.
- Ensured compliance with Health, Safety, and Environment (HSE) standards and worked closely with faculty advisors, resulting in an orderly program and a successful onboarding experience for participants.

HIMA TG "TERRA" ITB Election

Head of Disciplinary Commission

Bandung, Indonesia

December 2022 – May 2023

- Developed and enforced clear election rules and policies to ensure fair and transparent student association elections for a 200+ member organization.
- Supervised the electoral process, addressed and adjudicated any violations or disputes, and upheld disciplinary standards to maintain the integrity of the election.
- Recommended procedural improvements for future elections, contributing to a more robust and transparent electoral system.

OSKM ITB 2022

Lead Mentor

Bandung, Indonesia

July 2022 – August 2022

- Mentored and guided incoming freshmen during ITB's campus orientation, providing academic advice and personal support to ease their transition to university life.
- Coordinated a team of 80+ student mentors, leading training sessions and team-building activities to ensure a cohesive approach and an engaging orientation experience for over 3,000 new students.
- Fostered a welcoming and inclusive environment for newcomers, which strengthened community bonds and improved freshman participation in orientation events.

Education

Bandung Institute of Technology

Bachelor of Science in Geophysical Engineering, GPA: 3.30/4.0

Bandung, Indonesia

August 2025

- Honors: Dean's List (4th & 6th semesters for outstanding academic performance)

- Bachelor's Thesis: Sandstone Reservoir Characterization Using Seismic Inversion of Acoustic Impedance, Shear Impedance, and Extended Elastic Impedance (Talang Akar Formation, South Sumatra Basin)
- Relevant Coursework: Seismic Data Processing, Reservoir Geophysics, Petrophysics, Geostatistics, Signal Processing

Skills & Interest

Technical:

- Software & Tools: Petrel, HRS, Vista, ProMAX, SonarWiz, Oasis Montaj, Refrappy, Surfer, Google Earth Pro, QGIS, IPI2WIN, ModelVision, RES2D, Arduino IDE, Geolog, Interactive Petrophysics (IP).
- Programming & Data: Python (Jupyter Notebook), basic Linux/Unix and shell scripting, MATLAB (fundamentals), data analysis with libraries (NumPy/Pandas).
- Design & Media: Adobe Premiere Pro, Adobe Photoshop, Figma, Canva.
- Productivity: Microsoft Office Suite, Google Workspace, Notion.

Language:

- Indonesian: Native.
- English: Professional Working Proficiency.

Interest:

- Marine and offshore geophysical survey operations, including vessel-based data acquisition and field support
- Geophysical data processing, quality control, and integration for offshore site investigation and subsea projects
- Field engineering roles and on-site operations (e.g., Field Engineer)